

Lecture Notes: Sequences

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1 Convergence

Definition 1 (Convergence). A real sequence (a_n) converges to $L \in \mathbb{R}$ if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $|a_n - L| < \varepsilon$.

Example 2. The sequence $a_n = 1/n$ converges to zero. Given $\varepsilon > 0$, choose $N > 1/\varepsilon$; then $n \geq N$ implies $1/n < \varepsilon$.

Theorem 3 (Uniqueness of limits). *A convergent real sequence has exactly one limit.*

Proof. Suppose $a_n \rightarrow L$ and $a_n \rightarrow M$ with $L \neq M$. Set $\varepsilon = |L - M|/3$. For sufficiently large n ,

$$|L - M| \leq |L - a_n| + |a_n - M| < 2\varepsilon = \frac{2}{3}|L - M|,$$

a contradiction. Thus $L = M$. □

Remark. Theorem 3 justifies writing “the limit” of a convergent sequence.

2 A useful consequence

Proposition 4. *Every convergent real sequence is bounded.*

Proof. Suppose $a_n \rightarrow L$. Choose N so that $n \geq N$ implies $|a_n - L| < 1$, and set

$$B = \max\{|a_1|, \dots, |a_N|, |L| + 1\}.$$

For $n < N$ the bound $|a_n| \leq B$ follows from the definition of B . For $n \geq N$, the triangle inequality gives $|a_n| \leq |a_n - L| + |L| < |L| + 1 \leq B$. □